



Applied Statistical Analysis II

Replication Project

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••••• *The Paper*

“The Influence of Religious–Political Sophistication on U.S. Public Opinion”

Author: Eric R. Schmidt

Source: Political Behavior, Volume 40, pages 21–53, (2018)

Main idea: Understand the effect of RPS on the public opinion of religious identifiers



What is RPS?

Religious - Political Sophistication

“knowledge of formal church teaching”

(Schmidt, 2017)

i.e. For this study, a religious identifier’s knowledge of their church’s official stance on political issues

The Hypotheses

Hypothesis 1:

Religious identifiers who know their church's official teachings on these issues are more likely to share these positions.

Hypothesis 2:

For religious identifiers, RPS has a greater impact on public opinion than secular political knowledge.

Hypothesis 3:

RPS helps connect their faith to politics better than (neutral) religious facts.
Ex. scripture

Data

Source: the 2014 Cooperative Congressional Election Study (CCES) and an original CCES module

```
#subsetting the data to exclude ambiguous answers  
mainliners = subset(rep_data, mainline == 1 & ambiguous == 0)  
evan.cath = subset(rep_data, mainline != 1 & ambiguous==0)
```

- Best represented religious traditions
 - Roman Catholics (206)
 - Evangelical Protestants (158)
 - Mainline Protestants (97) *

Outcome:

Support for:

- Same-sex marriage (SSM) rights
- Abortion rights

Input variables:

Party ID (0 = strong Democrat; 6 = strong Republican)

Church attendance (0 = never; 5 = more than once per week)

Biblical Literacy (the number of correct answers to three multiple-choice questions about the Bible's books and figures)

Secular Political Knowledge (a three-point scale indicating each respondent's number of correct responses to two questions about national politics)

RPS (1 = respondent knows official denomination stance; 0 = respondent doesn't know)

Controlled variables: age, ideology, gender, race, education, family income, and religious tradition (1 = Roman Catholic; 0 = Evangelical Protestant)

Models

1. **Secular Model:** $\text{Support} = \alpha + \beta_1 \times \text{Party Identification} + \beta_2 \times \text{Secular Political Knowledge} + \beta_3 \times \text{Party ID} \times \text{Secular Political Knowledge}.$
2. **Naive Model:** $\text{Support} = \alpha + \beta_1 \times \text{Party Identification} + \beta_2 \times \text{Secular Political Knowledge} + \beta_3 \times \text{Party ID} \times \text{Secular Political Knowledge} + \beta_4 \times \text{Church Attendance} + \beta_5 \times \text{Biblical Literacy} + \beta_6 \times \text{Church Attendance} \times \text{Biblical Literacy}.$
3. **“Culture Wars” Model:** $\text{Support} = \alpha + \beta_1 \times \text{Party Identification} + \beta_2 \times \text{Secular Political Knowledge} + \beta_3 \times \text{Party ID} \times \text{Secular Political Knowledge} + \beta_4 \times \text{Church Attendance} + \beta_5 \times \text{Biblical Literacy} + \beta_6 \times \text{Church Attendance} \times \text{Biblical Literacy} + \beta_7 \times \text{Church Attendance} \times \text{Secular Political Knowledge}.$
4. **RPS Model:** $\text{Support} = \alpha + \beta_1 \times \text{Party Identification} + \beta_2 \times \text{Secular Political Knowledge} + \beta_3 \times \text{Party ID} \times \text{Secular Political Knowledge} + \beta_4 \times \text{Church Attendance} + \beta_5 \times \text{Biblical Literacy} + \beta_6 \times \text{Church Attendance} \times \text{Biblical Literacy} + \beta_7 \times \text{Church Attendance} \times \text{Secular Political Knowledge} + \beta_8 \times \text{RPS} + \beta_9 \times \text{Church Attendance} \times \text{RPS}.$

Example from the study

This table shows Evangelical Protestants and Roman Catholics’ support for same sex marriage with all four models.

Main findings: The church x rps(ssm or abortion) interaction term is significant.

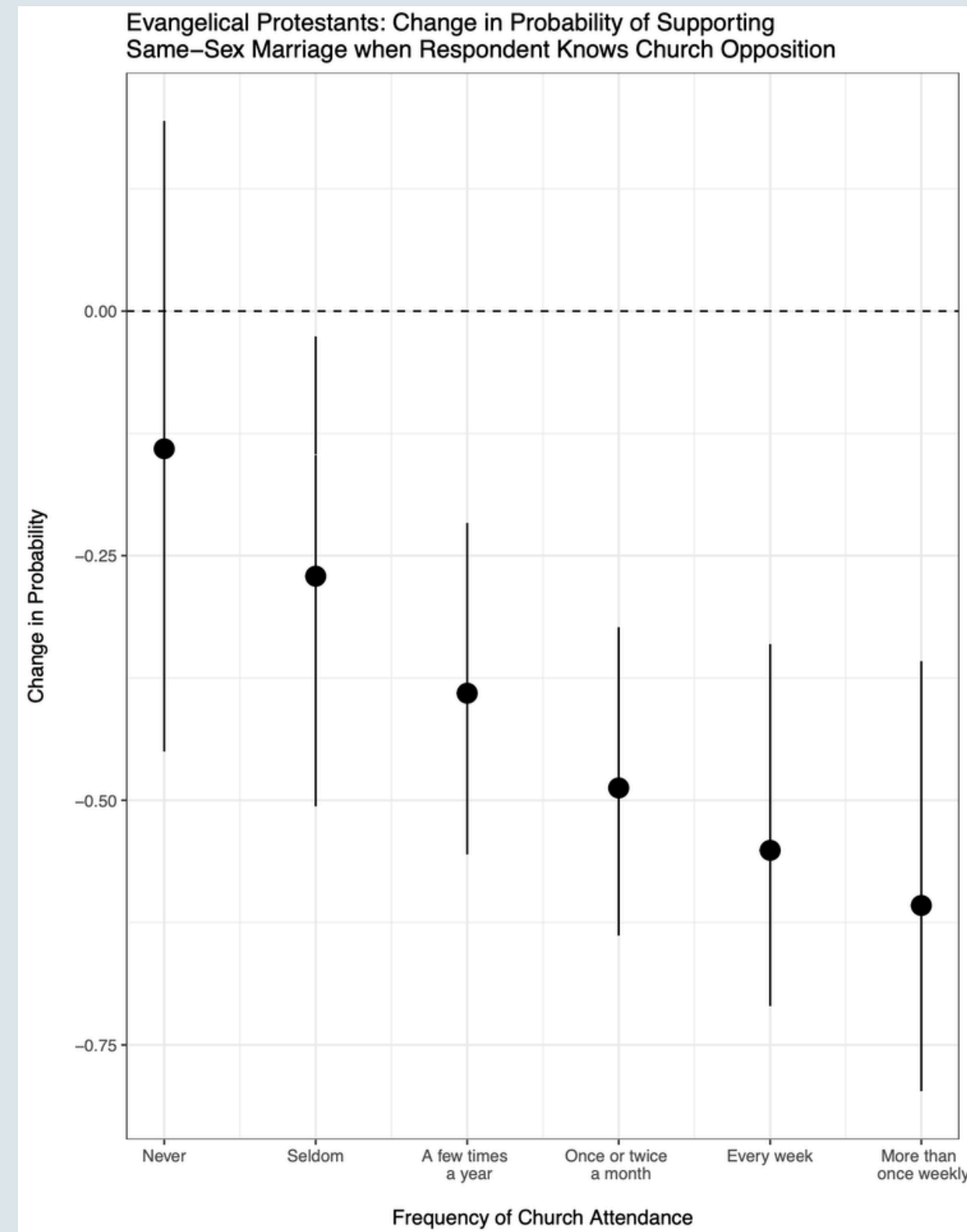
What does that mean?

Table 1 Support for same sex marriage: Evangelical Protestants and Roman Catholics					secular.knowledge	−0.082	−0.141	−0.108	0.040
						(0.308)	(0.354)	(0.471)	(0.519)
					partyid × secular.knowledge	−0.002	0.002	0.004	−0.078
						(0.088)	(0.098)	(0.099)	(0.116)
					church		−0.080	−0.061	0.044
							(0.263)	(0.322)	(0.378)
					bible.collapsed		−0.262	−0.261	−0.084
							(0.512)	(0.511)	(0.550)
					church × bible.collapsed		−0.074	−0.074	−0.054
							(0.171)	(0.170)	(0.185)
					secular.knowledge × church			−0.014	0.075
								(0.138)	(0.159)
					rps.ssm				−0.596
									(0.764)
					church × rps.ssm				−0.605*
									(0.262)
					Num.Obs.	280	246	246	246
					AIC	301.5	265.1	267.1	229.0
					BIC	341.5	314.2	319.7	288.6
					Log.Lik.	−139.734	−118.555	−118.549	−97.484
					RMSE	0.41	0.40	0.40	0.36
					+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

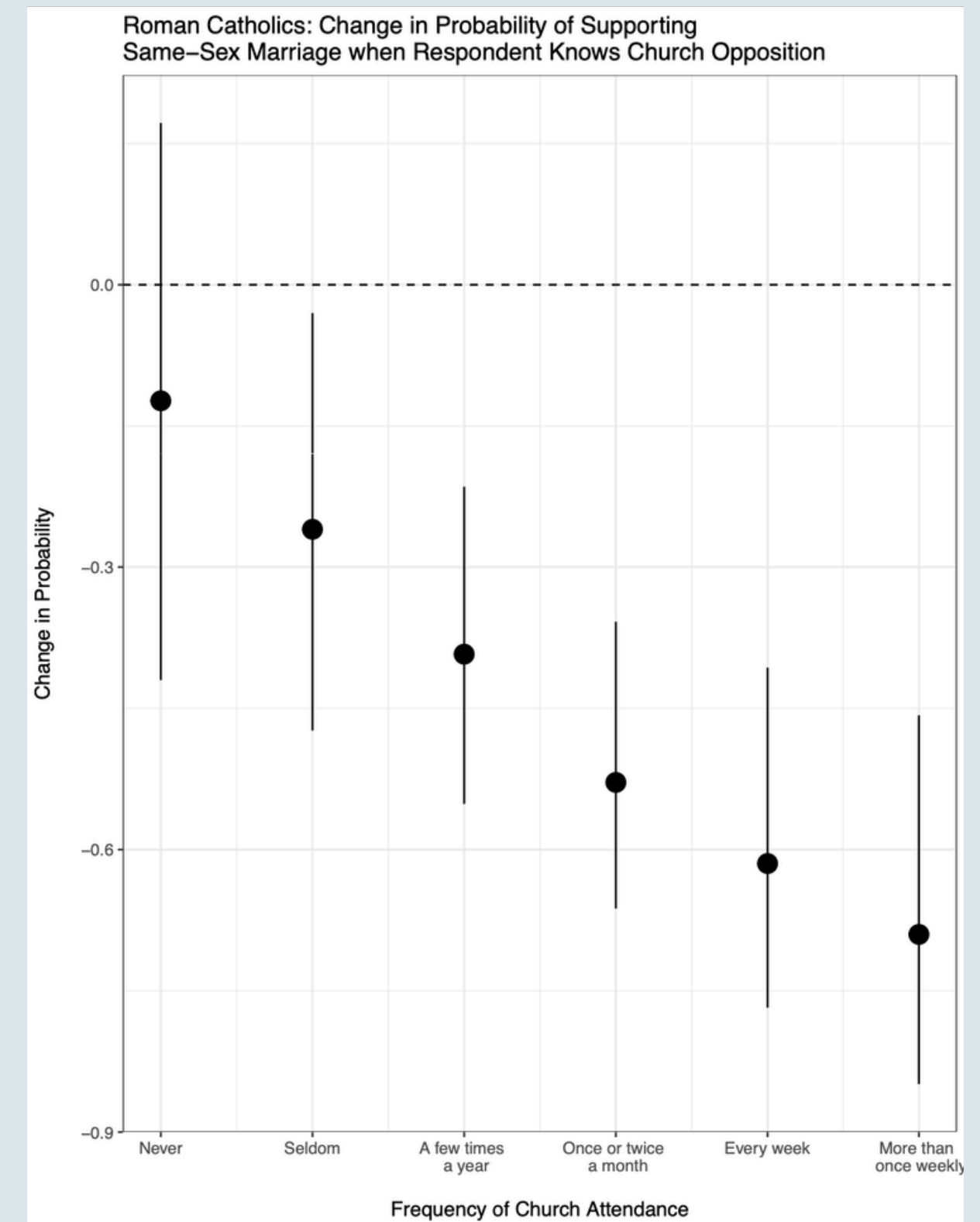
Replicated Figures 1 & 2

Evangelical Protestants

The figures show the predicted probability of when Evangelical Protestants and Roman Catholics know their church's stance on same sex marriage and align with the church's views when the frequency of church attendance increases.



Roman Catholics



My extension

Part 1: The Reasoning

3. “**Culture Wars**” Model: $\text{Support} = \alpha + \beta_1 \times \text{Party Identification} + \beta_2 \times \text{Secular Political Knowledge} + \beta_3 \times \text{Party ID} \times \text{Secular Political Knowledge} + \beta_4 \times \text{Church Attendance} + \beta_5 \times \text{Biblical Literacy} + \beta_6 \times \text{Church Attendance} \times \text{Biblical Literacy} + \beta_7 \times \text{Church Attendance} \times \text{Secular Political Knowledge}.$

Not significant in most models



- Expectation: “we should expect a negative and statistically significant coefficient on this interaction term” (Schmidt, 2017) - **NOT** fully met
- Testing whether the data actually provides evidence that this mechanism works how the theory predicts
- Looking at 3 out of 4 “Culture Wars” models
 - Evangelical Protestants and Roman Catholics’ support for SSM
 - Evangelical Protestants and Roman Catholics’ support for Abortion
 - Mainline Protestants’ support for SSM

My extension

Part 2: The Test +code

Null: The interaction has no effect on policy support.

1. re-run the full model
2. created and ran the reduced model
3. Ran Likelihood Ratio Test

*Repeated for the other models

```
#Evangelical and Roman Catholics: SSM
cw1 <- glm(ssm ~ age + female + black + education.collapsed
           + income.collapsed + ideology.collapsed + romancatholic
           + partyid + secular.knowledge + partyid*secular.knowledge
           + church + bible.collapsed + church*bible.collapsed
           + church*secular.knowledge,
           data=evan.cath, family="binomial")
summary(cw1)

cw1_reduced <- glm(ssm ~ age + female + black + education.collapsed
                  + income.collapsed + ideology.collapsed + romancatholic
                  + partyid + secular.knowledge + partyid*secular.knowledge
                  + church + bible.collapsed + church*bible.collapsed,
                  data=evan.cath, family="binomial")
summary(cw1)

evan.cath_ssm <- anova(cw1_reduced, cw1, test = "Chisq")
```

My extension

Part 3: The Results

- p-values > 0.05
- Fail to reject, no statistical evidence that the interaction between secular knowledge and church attendance improves model fit for these three models.

Table 9: Likelihood Ratio Test: Evangelical Protestants and Roman Catholics' support of SSM

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	232	237.110			
2	231	237.099	1	0.011	0.918

Table 10: Likelihood Ratio Test: Evangelical Protestants and Roman Catholics' support of Abortion

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	231	224.957			
2	230	224.957	1	0.001	0.980

Table 11: Likelihood Ratio Test: Mainline Protestants' support of Abortion

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	75	95.070			
2	74	94.731	1	0.339	0.561



Thank you!



References

Schmidt, Eric R. (2017). “The Influence of Religious–Political Sophistication on U.S. Public Opinion.” *Political Behavior*, 40(1), 21–53. <https://doi.org/10.1007/s11109-017-9390-z>